

Legal AI Assistant Design: A Comparative Study of Retrieval-Augmented Generation and Fine-Tuning on Belgian Statutory Law

Progress Report — June/July 2026 (Revised and Extended)

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Abstract

This report presents the revised experimental protocol and the extended set of results produced since our previous progress report. That earlier report compared a zero-shot Mistral-7B-Instruct baseline, a retrieval-augmented (RAG) configuration, and a QLoRA fine-tuned model on a 50-question subsample, and was flagged by the supervisor as insufficiently rigorous for a two-student project: comparisons appeared limited to a single run, extension to different data samples and subgroups of law was requested, and technical difficulties were to be documented explicitly. This report responds directly to that feedback. We evaluate four configurations (zero-shot, RAG, fine-tuning, and fine-tuning+RAG combined) on the full 222-question BSARD test split, report 95% bootstrap confidence intervals and Holm-Bonferroni corrected paired significance tests (bootstrap and Wilcoxon) for every comparison, replicate the fine-tuning run over three random seeds, decompose the RAG error budget between retrieval and generation via an oracle-context ceiling experiment, verify generalization on a second base model (Qwen2.5-7B-Instruct), test abstention behaviour on out-of-scope questions, operationalize the supervisor's subgroup suggestion through a Code Civil legal-domain specialist adapter, and complement automatic metrics with an independent LLM-judge and a three-annotator blind human evaluation arena (Cohen's Kappa). We find that citation grounding, not surface text quality, is the dominant axis of difference between configurations, that most of the residual error in RAG stems from retrieval failure rather than generation, that neither RAG nor fine-tuning corrects a systematic quality drop on questions requiring synthesis across four or more statutory articles, and that blind human judgment favours RAG over fine-tuning alone far more decisively than lexical overlap metrics from our earlier small-sample experiment suggested — a finding that itself illustrates why the extended protocol was necessary.

1. Introduction

The goal of this Master's thesis (TER) is to build and rigorously evaluate a French-language legal question-answering assistant, comparing two adaptation strategies for a small open-weight language model: retrieval-augmented generation (RAG) and parameter-efficient fine-tuning (QLoRA), on Mistral-7B-Instruct-v0.3 (Jiang et al., 2023). The original project target was French law via Légifrance/DILA, using the LLeQA benchmark (Louis et al., 2023); an access request for LLeQA remained unanswered since April 2026, and the project substituted BSARD (Louis and Spanakis,

2022), a statutory article retrieval dataset covering Belgian French-speaking law. This substitution is documented and assumed explicitly: the object of study is the comparison of adaptation methods under a shared, reproducible, access-free protocol, not the legal content of a specific jurisdiction, and the language (French) is preserved.

Three research questions structure the work throughout:

- RQ1 — Can a small (7B) open-weight model, specialized for a narrow domain, compete with an unspecialized model on a precise legal question-answering task?
- RQ2 — Under what conditions does fine-tuning yield a real gain over RAG alone, and does combining both help further?
- RQ3 — Does the resulting system cite its legal sources correctly, and does it recognize the limits of its own competence (abstention) rather than fabricate an answer?

In June/July, an initial progress report (submitted as chaab1.pdf) presented a first iteration of this comparison: a single fine-tuning run (LoRA rank 32, 580 training examples), a retrieval upgrade from an mpnet baseline to multilingual-e5-large, and a three-way comparison of zero-shot, RAG, and fine-tuned Mistral on a 50-question subsample. The supervisor's feedback on that report was explicit: a single experiment is not sufficient evidence for a two-student project; the protocol should be extended to test on different data samples, subgroups of law should be studied separately, and technical difficulties should be documented as an appendix rather than narrated inline. This report is the direct response to that feedback, produced under a compressed timeline (defense at end of August 2026) using free-tier compute (Kaggle) followed by a rented GPU (RunPod RTX 4090) once the experimental surface had grown beyond what free-tier quotas could support.

2. Related Work

Parameter-efficient fine-tuning is dominated by LoRA (Hu et al., 2021), which freezes the pretrained weight matrix and learns a low-rank update $W' = W_0 + BA$ with $B \in \mathbb{R}^{(d \times r)}$, $A \in \mathbb{R}^{(r \times k)}$, $r \ll \min(d, k)$; QLoRA (Dettmers et al., 2023) combines this with 4-bit NF4 quantization of the frozen backbone, which is the exact configuration used here. Retrieval-augmented generation follows the original formulation of Lewis et al. (2020), surveyed extensively by Gao et al. (2023); more recent work such as Self-RAG (Asai et al., 2023) revisits the naive retrieve-then-generate pipeline, which we use as-is, without adaptive retrieval or self-critique, as a deliberately simple and reproducible baseline.

The direct methodological reference for this work is Derby LLM (Bouvard, Ciancone, Gourru and Schaeffer, APIA@PFIA 2024), which compares RAG and fine-tuning on French-language enterprise data using an evaluation arena and a faithfulness metric; our protocol is positioned explicitly against theirs (Section 6.9 reproduces their faithfulness formulation; Section 7 reproduces their arena methodology, extended with confidence intervals and Cohen's Kappa, which Derby LLM does not report). Balaguer et al. (2024) study RAG vs. fine-tuning on an agricultural domain and report that combining both is often complementary; Ovadia et al. (2023) argue that fine-tuning is a comparatively poor vehicle for injecting new factual knowledge relative to retrieval. Our own oracle-ceiling

decomposition (Section 6.5) and seed-variance analysis (Section 6.2) engage directly with both claims. Evaluation of generative QA draws on BERTScore (Zhang et al., 2020), ROUGE-L (Lin, 2004), RAGAS (Es et al., 2023), and the Chatbot Arena human-preference methodology (Chiang et al., 2024). For legal NLP, BSARD (Louis and Spanakis, 2022) and LLeQA (Louis et al., 2023) are the direct dataset lineage; LegalBench (Guha et al., 2023), CUAD (Hendrycks et al., 2021) and LexGLUE (Chalkidis et al., 2022) situate the broader legal-NLP benchmark landscape, though none targets French/Belgian statutory retrieval specifically.

3. Dataset and Task

BSARD (Belgian Statutory Article Retrieval Dataset) contains 22,633 articles drawn from Belgian federal and regional legal codes, with 886 human-written questions (split 886 train / 222 test, each annotated with the gold article identifiers required to answer it) and an additional 113,165 unannotated synthetic paraphrase questions, never used prior to this report. Questions are natively categorized into 7 legal subject areas (Famille, Logement, Argent, Justice, Étrangers, Travail, Protection sociale), which we exploit directly for the subgroup analysis requested by the supervisor (Section 6.4).

A methodological limitation is stated explicitly rather than hidden: BSARD is natively a retrieval dataset and does not provide a gold free-text answer. The reference answer used for ROUGE-L and BERTScore is reconstructed by concatenating the text of the gold cited articles. Consequently, ROUGE-L/BERTScore should be read as a measure of lexical overlap with the statutory text, not of writing quality per se, and the low absolute values reported throughout this document (0.08–0.22) should be interpreted in that light rather than as a sign of catastrophic failure on their own.

4. Method

4.1 Four configurations

Four configurations are compared throughout, using an identical generation prompt template for fairness: C1 (zero-shot Mistral-7B-Instruct-v0.3, no context), C2 (RAG: base model with retrieved context), C3 (QLoRA fine-tuned model, no retrieval), C4 (fine-tuned model with retrieval). This is an extension of the three-configuration comparison in the previous report (which lacked C4); adding the combined configuration operationalizes the second research question directly.

4.2 Retrieval pipeline

Passage embeddings are computed with `intfloat/multilingual-e5-large` using the model's required “passage:” / “query:” prefixes; retrieval scores documents by cosine similarity $\text{sim}(\mathbf{q}, \mathbf{d}) = (\mathbf{q} \cdot \mathbf{d}) / (\|\mathbf{q}\| \cdot \|\mathbf{d}\|)$ against an index built once over the full 22,633-article corpus. Production configurations (C2, C4) use $\text{top-k} = 5$, chosen after the k -ablation reported in Section 6.7.

4.3 QLoRA fine-tuning

The base model is loaded in 4-bit NF4 (bitsandbytes), with a LoRA adapter of rank $r = 32$ applied to the attention projection matrices (an attention+MLP ablation is reported separately, Section 6.2). The main configuration trains on the 580 official BSARD

training questions paired with their reconstructed reference answers, formatted via the Mistral chat template, for 3 epochs. A held-out validation split of 100 questions (fixed seed 999, independent of the training seed, identical across every run reported in this document) was introduced in this revision specifically to detect overfitting, which the single earlier run could not do.

5. Experimental Setup — Addressing the “Single Experiment” Feedback

Concretely, the following elements were added or changed relative to the previous (chaabl.pdf) report, each directly traceable to a specific point of supervisor feedback:

- Full 222-question test split for every core metric, instead of a 50-question subsample.
- 95% confidence intervals via non-parametric bootstrap (1,000 resamples with replacement) on every reported mean.
- Paired significance testing between every configuration pair, using both the bootstrap paired-difference test and the Wilcoxon signed-rank test (which assumes a symmetric difference distribution; agreement between the two increases confidence in a conclusion), with Holm-Bonferroni correction across the 6 pairwise comparisons.
- Subgroup analysis by the 7 native BSARD legal categories — direct implementation of “you can focus on particular subgroups of laws”.
- A dedicated Code Civil legal-domain specialist adapter (trained on the 331 training questions touching that code only), compared in- and out-of-domain against the generalist adapter — a second, stronger implementation of the same subgroup suggestion.
- Three-seed replication (42, 123, 2026) of the main fine-tuning and generation pipeline, to report variance rather than a single point estimate.
- A second base model (Qwen2.5-7B-Instruct) run through the identical zero-shot/RAG protocol, to check whether findings generalize beyond Mistral.
- An oracle-context ceiling experiment isolating retrieval error from generation error.
- A 50-question abstention stress test (out-of-domain, foreign-law, and withdrawn-article questions) probing whether the system recognizes the limits of its competence.
- An independent LLM-judge (Phi-3.5-mini-instruct, chosen specifically to avoid the self-preference conflict of using Qwen2.5-7B as both a compared model and a judge) scoring relevance only, 10 stochastic samples per response, since fidelity is already measured deterministically (Section 6.9).
- A three-annotator blind human evaluation arena with Cohen's Kappa inter-annotator agreement, extending Derby LLM's own arena methodology, which reports neither confidence intervals nor inter-annotator agreement.

6. Results

6.1 Retrieval evaluation

Method	R@1	R@3	R@5	R@10	R@20	MRR@10	nDCG@10
BM25	0.063	0.140	0.184	0.224	0.289	0.215	0.176
mpnet	0.021	0.045	0.074	0.097	0.124	0.105	0.076
multilingual-e5-large	0.083	0.169	0.211	0.296	0.377	0.288	0.230
Hybrid BM25+e5 (RRF)	0.106	0.196	0.250	0.322	0.390	0.312	0.258

Table 1 — Retrieval quality on the full 222-question test split, 22,633-article corpus.

Reconciliation with the previous progress report

The previous report (chaab1.pdf) stated Recall@10 = 34.7% for mpnet and 52.3% for multilingual-e5-large, roughly 2–2.5× higher than the values reproduced here (9.7% and 29.6% respectively) under an identical stated protocol (same corpus, same models, correct e5 prefixes verified). This gap is presented here as an open, unresolved question rather than papered over. The article-length distribution has been checked (median 77 words; only 6.8% exceed 384 words, mpnet's truncation limit), which rules out truncation as the sole explanation for a gap of this magnitude. Two hypotheses remain untested at time of writing: (i) the previous indexing pass may have included hierarchical metadata (code/chapter/section headers) alongside the article body, adding lexical signal absent from the present body-only index; (ii) the previous retriever may have used a lightweight contrastive fine-tune on the BM25 hard negatives BSARD provides, rather than an off-the-shelf embedding model. This will be investigated via a chunking/document-enrichment ablation before the final report is written; we report it here, rather than silently adopting either number, precisely because a single unverified figure carried forward without a reproducibility check is the exact failure mode this report exists to correct.

6.2 Fine-tuning ablations and seed variance

Run	Seed	n_train	n_val	r	Target	Train loss
seed123_n580_r32_attn	123	580	100	32	attn	0.844
seed2026_n580_r32_attn	2026	580	100	32	attn	0.850
seed42_n190_r32_attn	42	190	100	32	attn	1.122
seed42_n331_r32_attn_CodeCivil	42	331	100	32	attn	0.832
seed42_n380_r32_attn	42	380	100	32	attn	0.966
seed42_n580_r32_attn (main)	42	580	100	32	attn	0.834
seed42_n580_r32_attn_mlp	42	580	100	32	attn+MLP	0.527
seed42_n580_r32_attn_synth	42	2580	100	32	attn	0.940

Run	Seed	n_train	n_val	r	Target	Train loss
seed42_n580_r8_attn	42	580	100	8	attn	0.850
seed42_n786_r32_attn	42	786	100	32	attn	0.736

Table 2 — Ten fine-tuning ablation runs. “Overfit gap” = eval loss – train loss.

Across the three seeds of the main configuration (42, 123, 2026; r=32, attn, n=580), final training loss is 0.8426 ± 0.0063 — low variance on the training objective itself. However, downstream generation quality varies considerably more across seeds (Table 3), and adding attention+MLP targets or synthetic data both reduce eval loss substantially (0.416 and 0.516 respectively) at the cost of a markedly larger overfit gap (+0.214 and −0.235 — the negative gap for the synthetic-augmented run indicates eval loss below train loss, itself a symptom of distribution mismatch between the noisier synthetic training pairs and the clean validation set, not of good generalization).

Reconciliation with the previous report's headline fine-tuning numbers

The previous report's second iteration (also r=32, n=580, seed unspecified) reported ROUGE-L = 0.1612 and BERTScore F1 = 0.7043, measured on a 50-question subsample. Under the present protocol (seed 42, full 222-question test split), the identical configuration yields ROUGE-L = 0.1467, with a 95% bootstrap CI of [0.1360, 0.1598] (Table 4). The earlier point estimate falls just outside the upper bound of this interval. This is not treated as a contradiction to resolve but as a direct, concrete illustration of exactly the risk the supervisor's feedback warned against: a single 50-question estimate is close to, yet measurably different from, the full-test-split estimate purely due to sampling variance, which is precisely why every quantity in this report is now accompanied by a confidence interval rather than reported as a lone number.

6.3 Generation quality across four configurations

Config	ROUGE-L	BERTScore F1	Cit. precision	Cit. recall	Exact match	Hallucination
C1 zero-shot	0.110	0.779	0.016	0.011	0.005	0.144
C2 RAG	0.140	0.795	0.176	0.132	0.036	0.234
C3 fine-tune	0.147	0.801	0.023	0.012	0.000	0.050
C4 fine-tune+RAG	0.179	0.798	0.069	0.053	0.005	0.167

Table 3 — Full test split (n=222), single-seed (42) point estimates.

Config	ROUGE-L mean	95% CI low	95% CI high
C1 zero-shot	0.1100	0.1037	0.1163
C2 RAG	0.1395	0.1292	0.1511

Config	ROUGE-L mean	95% CI low	95% CI high
C3 fine-tune	0.1467	0.1360	0.1598
C4 fine-tune+RAG	0.1790	0.1565	0.2017

Table 4 — Bootstrap 95% confidence intervals, 1,000 resamples.

Comparison	Δ ROUGE-L	p (bootstrap)	p (Wilcoxon)	Significant (Holm)	Tests agree
C1 vs C2	−0.0295	<0.0001	<0.0001	yes	yes
C1 vs C3	−0.0368	<0.0001	<0.0001	yes	yes
C1 vs C4	−0.0691	<0.0001	<0.0001	yes	yes
C2 vs C3	−0.0072	0.296	0.080	no	yes
C2 vs C4	−0.0395	<0.0001	0.0002	yes	yes
C3 vs C4	−0.0323	0.004	0.307	yes	no

Table 5 — Paired significance tests, Holm-Bonferroni corrected across 6 comparisons. Both configurations that add RAG or fine-tuning individually beat zero-shot decisively; C2 vs. C3 (RAG alone vs. fine-tuning alone) is the only pair both tests agree is not significant; C3 vs. C4 is the only pair where the two tests disagree, treated as a genuinely borderline result rather than resolved in favour of one test.

6.4 Category-level subgroup analysis

Category	n	C1	C2	C3	C4
Argent	36	0.112	0.167	0.128	0.191
Étrangers	13	0.096	0.118	0.126	0.117
Famille	67	0.117	0.146	0.179	0.182
Justice	30	0.082	0.099	0.105	0.116
Logement	66	0.118	0.140	0.149	0.216
Protection sociale	4	0.087	0.126	0.112	0.137
Travail	6	0.121	0.153	0.154	0.138

Table 6 — ROUGE-L by legal category (direct implementation of “focus on subgroups of laws”). Protection sociale (n=4) and Travail (n=6) are flagged explicitly as underpowered for any individual-category claim and are reported for completeness, not as reliable estimates.

6.5 Oracle-context ceiling: decomposing retrieval error from generation error

To determine whether RAG’s residual error originates in retrieval (wrong articles found) or in generation (right articles found, poor use of them), the true gold articles are injected directly into the prompt, bypassing the retriever entirely, and compared against C1 (no

context) and C2 (real retrieval).

Condition	ROUGE-L	BERTScore	Cit. precision	Cit. recall	Exact match	Hallucination
C1 zero-shot (no context)	0.110	0.779	0.016	0.011	0.0045	0.144
C2 RAG (real retrieval)	0.140	0.795	0.176	0.132	0.0360	0.234
Oracle context (gold articles)	0.252	0.835	0.665	0.544	0.3108	0.117

Table 7 — Oracle-context ceiling. Base model (non-fine-tuned) throughout.

The gap in exact-match citation rate between real retrieval and the oracle ceiling is 27.5 percentage points (3.6% $\rightarrow$ 31.1%), against a gap of only 0.5 points between real retrieval and no context at all (0.45% $\rightarrow$ 3.6%). Nearly the entire achievable gain from RAG is therefore gated by retrieval quality, not by the generator's ability to use context once it is correct — retrieval, not generation, is the dominant bottleneck of the RAG pipeline as currently configured. A second, counter-intuitive finding is that hallucination rate is higher under real retrieval (0.234) than under no context at all (0.144): injecting a plausible but wrong article appears to invite more confident fabrication than injecting nothing, worse even than the oracle condition (0.117). This nuances any claim that “more context is strictly safer” and is discussed further in Section 8.

6.6 Cross-model generalization: Qwen2.5-7B-Instruct

Config (Qwen2.5-7B)	ROUGE-L	BERTScore	Cit. exact match	Hallucination
Zero-shot	0.105	0.779	0.000	0.180
RAG	0.131	0.785	0.027	0.360

Table 8 — Second base model, identical protocol, $n=222$.

The same qualitative pattern observed with Mistral — RAG improves lexical overlap and citation accuracy but roughly doubles the hallucination rate relative to zero-shot (0.180 $\rightarrow$ 0.360) — reproduces on an architecturally distinct 7B model. This indicates the retrieval-induced hallucination effect documented in Section 6.5 is not a Mistral-specific quirk.

6.7 Retrieval depth (k) ablation

k	ROUGE-L	BERTScore	Cit. precision	Cit. recall	Exact match	Hallucination	Avg. context (chars)
1	0.135	0.795	0.133	0.075	0.02	0.14	1,170
3	0.150	0.799	0.190	0.152	0.04	0.24	3,531
10	0.136	0.792	0.114	0.078	0.02	0.20	12,141

Table 9 — *k*-ablation, seed 42, *n*=50 (subsample used for this specific ablation).

Quality peaks at *k*=3 on every metric; *k*=10 does not improve citation accuracy over *k*=1 despite an order of magnitude more context (12,141 vs. 1,170 characters on average), consistent with a lost-in-the-middle-style dilution effect. The production choice of *k*=5 sits reasonably close to this empirical optimum.

6.8 Abstention on out-of-scope questions

A 50-question stress test (20 non-legal questions, 15 foreign-law questions, 15 questions referencing an article withdrawn from the index) probes whether the system recognizes when it should decline to answer rather than fabricate.

Config	Correct abstention (overall)	Non-legal	Foreign law	Withdrawn article
C1 zero-shot	0.0%	0.0%	0.0%	0.0%
C2 RAG	2.0%	0.0%	6.7%	0.0%

Table 10 — *Correct abstention rate by question type.*

The system essentially never declines to answer, even for a question about “the recipe for gaufres de Liège” or a jurisdiction it has no basis to discuss — it fabricates a plausible-sounding legal answer, including invented article numbers, in nearly every case. This directly and negatively addresses RQ3.

6.9 Fidelity (Derby LLM methodology)

Following Derby LLM’s explicit methodological choice — use an LLM judge only for what is intrinsically subjective (relevance), and a deterministic rule for what can be measured deterministically (factual grounding) — we reimplement their faithfulness metric rather than duplicate a single judgment two different ways. “Interest passages” (named entities, numbers, e-mail addresses, URLs, extracted with spaCy’s French NER pipeline) are identified in both the generated response and the gold reference fragment; fidelity is their overlap:

$$\text{fidelity}(\text{response}) = |\text{PI}^{\text{response}} \cap \text{PI}^{\text{fragment}}| / |\text{PI}^{\text{response}}|$$

Config	Mean fidelity	n with interest passages
C1 zero-shot	0.054	221/222
C2 RAG	0.295	219/222

Config	Mean fidelity	n with interest passages
C3 fine-tune	0.298	222/222
C4 fine-tune+RAG	0.324	210/222

Table 11 — Fidelity scores. Note Derby LLM does not report a bootstrap CI or significance test on this metric; we do so in the companion statistical appendix of the full thesis report.

6.10 LLM-as-judge: relevance

Following the same Derby-LLM-inspired separation of concerns, relevance (does the answer address what was asked, independent of factual correctness) is scored by an independent judge model, Phi-3.5-mini-instruct — chosen specifically because Qwen2.5-7B, used elsewhere as a second base model, would otherwise create a self-preference conflict. On a 50-question subsample and 4 configurations, each response is scored 10 times at temperature 0.7 (a single greedy judgment is a noisy point estimate for a non-deterministic judge) and averaged.

Config	Mean relevance (/5)	Mean intra-response SD	% scored 5/5	% scored $\leq 3/5$
C1 zero-shot	5.000	0.000	100.0%	0.0%
C2 RAG	4.850	0.048	90.2%	2.7%
C3 fine-tune	4.731	0.121	86.3%	3.3%
C4 fine-tune+RAG	4.617	0.107	77.1%	4.6%

Table 12 — Relevance scores and full-score distribution.

Mean relevance stays high throughout (4.6–5.0/5), but the distribution reveals a monotonically growing minority of clearly off-topic answers from C1 to C4 (0% → 4.6% scored $\leq 3/5$). Combined with the human arena qualitative findings (Section 7), this indicates that adding RAG or fine-tuning occasionally derails the answer from the specific question asked — a comparatively minor effect next to the citation-grounding gap documented in Sections 6.3 and 6.5.

6.11 Citation identifier format and difficulty correlates

BSARD’s article identifiers are heterogeneous across the 22,633-article corpus (13,199 “structured”, e.g. Art. L1122-9, vs. 9,434 “numeric simple”, e.g. Art. 959), discovered while debugging a citation-extraction regex (Appendix A.1). A Mann-Whitney test per configuration (118 simple vs. 104 structured test questions) finds no significant difference in citation exact-match rate between the two identifier formats in any of the four configurations (all $p > 0.20$), so this axis, while worth checking, does not explain performance variation on its own.

A Spearman correlation between question difficulty proxies and ROUGE-L, computed separately for every configuration, gives a sharper answer to the question “does the model behave uniformly across question difficulty?”:

Config	ρ (question length)	p	ρ (n gold articles)	p
C1 zero-shot	−0.107	0.113	−0.452	<0.0001
C2 RAG	−0.076	0.258	−0.336	<0.0001
C3 fine-tune	−0.064	0.340	−0.510	<0.0001
C4 fine-tune+RAG	−0.152	0.024	−0.353	<0.0001

Table 13 — Spearman ρ between question difficulty proxies and ROUGE-L.

Raw question length has little to no effect (non-significant in 3 of 4 configurations); the number of gold articles required to answer correctly is a strong, highly significant negative correlate in all four configurations. Grouping questions by this count confirms a sharp, structural quality drop once 4 or more articles must be synthesized (mean ROUGE-L falls from roughly 0.12–0.22 on 1–3-article questions to 0.07–0.11 on 4+-article questions, across every configuration, $n=83/76/63$ respectively). Neither RAG nor fine-tuning corrects this: multi-article synthesis remains a structural limitation of the current pipeline, not a retrieval or training deficiency specifically.

6.12 Predicting response reliability from observable features

A supervised classifier (logistic regression and random forest, 5-fold stratified cross-validation) is trained to predict whether a given response will achieve an exact-match citation, using only features available before generation (question length, number of gold articles, category training frequency, identifier format, configuration one-hot). A companion regressor predicts ROUGE-L directly.

Model	AUC (citation exact-match classifier)
Logistic regression	0.878
Random forest	0.838
Majority-class baseline	0.500

Model	R ² (ROUGE-L regressor)
Linear regression	0.136
Random forest	0.166

Table 14 — Reliability prediction. $n=888$ ($222 \text{ questions} \times 4 \text{ configurations}$), positive rate 1.13% for exact match (interpret AUC with the class imbalance in mind).

Random-forest feature importance ranks question length (0.389), category training frequency (0.185), and the number of gold articles (0.158) as the three strongest predictors of citation reliability — all available before generation, meaning a simple pre-flight classifier could plausibly flag low-confidence questions for human review in a deployed system, a practical implication we intend to explore further.

6.13 Specialist vs. generalist fine-tuning (Code Civil subgroup)

A dedicated adapter trained only on the 331 training questions touching the Code Civil is compared against the generalist adapter (580 examples, all domains), split by whether the 222 test questions themselves touch the Code Civil (n=86, in-domain) or not (n=136, out-of-domain).

Config	Subset	n	Generalist ROUGE-L	Specialist ROUGE-L	Generalist halluc.	Specialist halluc.
C3 fine-tune	In-domain	86	0.171	0.181	0.047	0.151
C3 fine-tune	Out-of-domain	136	0.132	0.123	0.052	0.022
C4 fine-tune+RAG	In-domain	86	0.215	0.151	0.151	0.081
C4 fine-tune+RAG	Out-of-domain	136	0.157	0.135	0.177	0.103

Table 15 — Specialist (n=331 train) vs. generalist (n=580 train) adapter, Code Civil subgroup.

The result is deliberately reported as mixed rather than framed as a clean win. Under C3, the specialist is marginally better on in-domain ROUGE-L but hallucinates three times more on its own specialty (0.151 vs. 0.047); under C4, the specialist underperforms the generalist on ROUGE-L in both domains, though it hallucinates less in both. An important confound must be stated explicitly: the specialist was trained on 331 examples against the generalist's 580, so this comparison conflates domain specialization with a reduction in total training volume, and the learning-curve results (Section 6.2, Table 2) already show that fewer examples alone degrade quality. We do not conclude from this experiment that domain specialization fails; we conclude that this particular experiment does not demonstrate it helps, and flag the confound for a properly controlled follow-up (matched training-set size) before the final report.

7. Human Evaluation Arena

Following Derby LLM's arena methodology (extended here with confidence-interval-style rigor absent from their original protocol), three annotators (Bartosz Konior, Chaabane Ouammou, and a third annotator, Mehdi, currently in progress) independently and blindly compare C2 (RAG) against C3 (fine-tuning), side A/side B randomized per question, with options “A is better”, “B is better”, “tie”, or “neither (both wrong)”, over 60 questions sampled with balanced category representation.

Annotator	C2 RAG preferred	C3 fine-tune preferred	Tie	Neither
Bartosz	42 (70.0%)	6 (10.0%)	5 (8.3%)	7 (11.7%)
Chaabane	33 (55.0%)	5 (8.3%)	9 (15.0%)	13 (21.7%)

Table 16 — Blind pairwise preference, C2 vs. C3, n=60 questions common to both annotators.

Cohen's Kappa on the 60 questions judged by both annotators is 0.389 (raw agreement 56.7%), a fair-to-moderate level of inter-annotator agreement, itself worth reporting honestly rather than assuming annotators trained on the same instructions will agree strongly — a limitation discussed further in Section 9. Despite this moderate agreement, both annotators independently and substantially prefer RAG over fine-tuning alone (70.0% and 55.0% respectively, against only 10.0% and 8.3% preferring the fine-tuned model). This stands in marked tension with the previous progress report's own automatic-metric comparison, which found the fine-tuned model ahead of RAG on both ROUGE-L (0.161 vs. 0.154) and BERTScore (0.704 vs. 0.659) on a 50-question subsample. We read this tension as informative rather than contradictory: it is a direct demonstration that a small-sample lexical-overlap comparison and a blind human-preference judgment on the same underlying systems can disagree considerably, which is itself the strongest argument in this report for the extended, multi-method protocol the supervisor requested.

8. Qualitative Error Analysis

Manual inspection by the two lead annotators (formal double-annotation with Cohen's Kappa, per the 9-category error taxonomy in `src/error_analysis.py`, is in progress and will be completed for the final report) surfaces four recurring failure patterns, each with a plausible mechanistic explanation grounded in the quantitative results above:

- Major legal contresens (role reversal): obligations are attributed to the wrong party (e.g. tenant vs. landlord, legal cohabitant vs. spouse). BSARD's synthetic question templates produce near-duplicate phrasings across adjacent categories (e.g. “married” vs. “legal cohabitation” variants of the same landlord-notice question), where the correct governing article differs subtly (direct application vs. “application by analogy”) — exactly the kind of confusable near-duplicate pair that both a dense retriever and a generator are prone to conflate.
- Hallucinated or invented statutory text: consistent with the near-zero abstention rate (Section 6.8) and the fact that hallucination remains at 11.7% even under the oracle context ceiling (Section 6.5), fabrication is not purely a retrieval artifact; some amount is a structural tendency of the base model to complete a plausible-sounding legal register regardless of whether it has grounding to do so.
- Complete topical drift: directly consistent with the LLM-judge relevance distribution (Section 6.10), where a growing minority of responses (up to 4.6% under C4) is scored as clearly off-topic; a wrong-but-adjacent retrieved passage appears to be a common trigger, since the generator has no mechanism to recognize that retrieved context does not in fact answer the question asked.
- Decoding artifacts (truncation, repetition loops): traced to a configuration bug, not a model or data limitation — generation used purely greedy decoding (temperature 0) without a repetition penalty or no-repeat n-gram constraint, a known cause of degenerate repetition loops on smaller models producing long structured output; correcting this is a concrete, low-cost action item for the final experimental pass.

A further limitation, raised independently by the annotators themselves, is worth stating plainly: neither is a trained jurist, and distinguishing a subtly wrong legal claim from a correct one (e.g. “applies directly” vs. “applies by analogy”) can exceed what a non-specialist can verify confidently from the shortened reference context shown in the arena interface. This is documented here as an explicit methodological limitation rather than smoothed over, and is one reason the moderate Cohen's Kappa (0.389) should not be over-interpreted as annotator carelessness.

9. Discussion

RQ1 — Can a specialized small model compete with an unspecialized one?

Partially. Fine-tuning alone (C3) beats zero-shot (C1) significantly on ROUGE-L ($p < 0.0001$) and nearly eliminates hallucination relative to every other configuration (0.050 vs. 0.144–0.234), but its citation exact-match rate remains effectively at zero (0.000), meaning the 7B model with only 580–786 training examples learns to write in a plausible legal register without reliably grounding its answers in the correct specific article — a distinction a purely lexical metric such as ROUGE-L does not surface on its own, and one only visible once citation-level metrics and the oracle-ceiling decomposition are examined jointly.

RQ2 — When does fine-tuning help, and does combining it with RAG help further?

Fine-tuning and RAG are not significantly different from each other on ROUGE-L (C2 vs. C3, $p = 0.296/0.080$, Table 5) but improve different things: RAG improves citation grounding far more (exact match 0.036 vs. 0.000) while fine-tuning nearly eliminates hallucination (0.050 vs. 0.234). Combining both (C4) gives the best point estimate on ROUGE-L and the category-level table (Table 6), but this is not fully stable: across the three-seed replication (Section 6.2), C4 beats C3 in only two of three seeds, and the C3-vs-C4 significance test is the one pair where the bootstrap and Wilcoxon tests disagree (Table 5) — so “combining helps”, as suggested by Balaguer et al. (2024), is only a partially and provisionally supported conclusion here, not a settled one. The oracle-ceiling result (Section 6.5) further suggests that most of the achievable improvement from adding RAG to a fine-tuned model is currently gated by retrieval quality rather than by the fine-tuned model's ability to use good context, aligning with Ovidia et al.'s (2023) broader claim that retrieval, not fine-tuning, is the more effective vehicle for injecting reliable factual grounding — though our fine-tuning advantage on hallucination suppression is a genuine counterpoint their framing does not fully anticipate.

RQ3 — Does the system cite correctly and recognize its own limits?

No, on both counts, and this is the central negative finding of the study. Citation exact-match never exceeds 3.6% in any real (non-oracle) configuration, and correct abstention on clearly out-of-scope questions is 0–2%. This mirrors Derby LLM's (Bouvard et al., 2024) own conclusion that RAG is more relevant than fine-tuning alone but that overall answer quality remains poor in either case on a comparably small real-world corpus — our results reach the same qualitative conclusion independently, on a different domain, language variety, and evaluation protocol, which we read as a modest but genuine corroboration rather than a coincidence attributable to shared methodology alone.

10. Limitations

- BSARD covers Belgian, not French, statutory law; the substitution is methodologically justified (Section 3) but the legal content itself cannot be claimed representative of French law.
- Reference answers for ROUGE-L/BERTScore are reconstructed from cited article text, not independently authored gold answers (Section 3).
- The unresolved retrieval Recall@k discrepancy against the previous report (Section 6.1) remains open at time of writing.
- The Code Civil specialist-vs-generalist comparison (Section 6.13) confounds domain specialization with training-set size.
- Human annotators are not trained jurists, which caps the precision of any individual arena judgment on subtly incorrect legal claims (Section 8); the third annotator's data collection is still in progress.
- Compute constraints mean some ablations (k-ablation, LLM-judge) use a 50-question subsample rather than the full 222; this is stated explicitly wherever it applies rather than left implicit.

11. Conclusion and Next Steps

This revision replaces a single 50-question, three-configuration comparison with a four-configuration, 222-question protocol carrying confidence intervals, corrected significance tests, subgroup analysis, multi-seed replication, a second base model, an oracle-ceiling decomposition, an abstention stress test, an independent LLM-judge, and a three-annotator human arena with measured inter-annotator agreement — directly and comprehensively answering the supervisor's June/July feedback. The central finding carried forward to the final report is that citation grounding, not lexical fluency, is the axis on which these systems actually fail, that this failure is dominated by retrieval error rather than generation error, and that neither technique nor their combination yet solves multi-article legal synthesis or abstention reliably. Remaining work before the end-of-August defense: complete the third annotator's arena data and the formal double-annotated error taxonomy; resolve or bound the retrieval Recall@k discrepancy via the planned chunking/enrichment ablation; re-run generation with a corrected decoding configuration (repetition penalty) to isolate how much of the qualitative error analysis is a fixable artifact; and finalize the LaTeX thesis report incorporating all results in this document.

Appendix A — Technical Difficulties and Practical Recommendations

Per the supervisor's explicit request, this appendix is written as practical library/tooling recommendations rather than a narrative account. Full detail (15 entries) is maintained in `DIFFICULTES.md` in the project repository; the most consequential are summarized here.

- Kaggle silently assigns a Tesla P100 GPU (CUDA sm_60) incompatible with current PyTorch builds unless `machine_shape` is set explicitly to `NvidiaTeslaT4` in `kernel-metadata.json` — the failure only surfaces after several minutes of setup, not at push time.

- BSARD's `article_ids` field is a comma-separated string, not a list; comparing it directly against integer IDs silently yields a `Recall@k` near zero with no error raised. Always assert the parsed type immediately after loading external data.
- BSARD article identifiers are not uniformly numeric (~32% use structured regional formats, e.g. Art. L1122-9, Art. R.II.21-7); a generalized regex with an explicit coverage assertion (`match_rate > 0.99`) is required before any citation metric can be trusted, and roughly 2.5% of identifiers carry a scraping-artifact suffix (e.g. “1714bis_REGION_DE_BRUXELLES-CAPITALE”) that must be stripped before use as a fine-tuning target or citation gold value.
- `trl`'s `SFTTrainer` has migrated training arguments into a dedicated `SFTConfig` class and introduced a chunked cross-entropy loss default (`loss_type="chunked_nll"`) that breaks on a 4-bit-quantized, PEFT-wrapped model; `loss_type="nll"` should be forced explicitly for any QLoRA setup on a fast-moving library version.
- Independent GPU jobs without local `src/` package imports (a deliberate deployment constraint) mean a bug fix in one script (e.g. incremental result saving after each ablation step, to survive an out-of-memory crash) does not propagate automatically to sibling scripts; a checklist of “universal” fixes should be reapplied manually across the job family.
- Kaggle's free-tier 2-simultaneous-GPU-session quota is consumed silently by browser-interactive sessions in addition to API-launched kernels, and re-pushing a kernel to fix a bug does not stop the previous, still-running version — both count against the quota simultaneously until the earlier version is stopped manually from the web interface.
- No held-out validation split existed for fine-tuning prior to this revision, making it impossible to distinguish memorization from generalization on 580–886 examples; a small ($n=100$), seed-fixed, cross-run-identical validation split with per-epoch evaluation was added specifically to enable this comparison.

Appendix B — Reproducibility Details

Component	Configuration
Base model	mistralai/Mistral-7B-Instruct-v0.3 (unsloth 4-bit mirror)
Second base model	Qwen2.5-7B-Instruct
Retriever	intfloat/multilingual-e5-large, cosine similarity, top-k=5
Quantization	4-bit NF4 (bitsandbytes)
LoRA rank (main)	$r=32$, $\alpha=64$, attention projections only
Fine-tuning data (main)	580 official BSARD training questions, 3 epochs
Validation split	100 questions, fixed seed 999, identical across all runs
LLM-judge	microsoft/Phi-3.5-mini-instruct, 10 samples/response, $T=0.7$
Fidelity NER	spaCy <code>fr_core_news_sm</code>
Bootstrap	1,000 resamples, percentile 95% CI

Component	Configuration
Multiple-comparison correction	Holm-Bonferroni, 6 pairwise comparisons
Compute	Kaggle T4 (free tier) → RunPod RTX 4090 (rented) for the extended wave

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